

ARUNAVA NAG

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SUMMARY

Data-driven researcher with extensive experience in statistical modeling, machine learning, and deep learning. Proficient in handling large (millions of rows) datasets, developing predictive models, and applying statistical methods to real-world problems. Adept at using Python, C++, and various data analysis/MLOps tools to drive insights and optimize performance.

TECHNICAL SKILLS

- **Languages:** Python, C++
- **Statistical Tools and Libraries:** Pandas, Pyspark, numpy, scipy, R
- **Software Framework:** Pytorch, pandas, tensorflow, CUDA, ROS
- **Concepts:** Statistical Modeling, Regression Analysis, Bayesian Networks, Deep Learning, Machine Learning, Neural Networks, Robotics, Data Visualization

WORK EXPERIENCE

Research Assistant, University of Nevada, Reno (Jan 2020 – Current)

- Developed and implemented statistical models and Kalman filtering techniques to enhance estimators for real-world dynamical systems on 15 million rows of real-world data from multiple sensors, resulting in an 82% prediction accuracy.
- Created a realistic simulator using probabilistic methods such as Gaussian processes and Bayesian optimization, significantly improving the robustness of predictions based on data driven approaches using real world statistics.

Senior Research Engineer, ROS-I AP, ARTC (Sep 2017- Dec 2019)

- Led a project integrating virtual reality with real world robot and applied computer vision techniques and virtual space to real world space mapping, and improved operator learning efficiency by 85%. Skills: Computer vision, deep learning, YOLO, virtual reality, Python, C#, C++.

Robotics Software Engineer, HOPE Technik (Feb 2017 – July 2017)

- Enhanced the pick-and-place accuracy for autonomous vehicles with 1 mm tolerance using computer vision techniques
- Collaborated with a cross-functional team to implement data-driven solutions for robot navigation, resulting in a 20% reduction in operational errors. Skills: Sensor fusion, computer vision, C++.

EDUCATION

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| • University of Nevada, Reno , PhD, Computer Science (est. 2024) | 3.75/4.0 |
| • North Carolina State University , MS, Electrical Engineering (2016) | 3.2/4.0 |
| • Visvesvaraya Technological University , BE, ECE (2013) | 3.5/4.0 |

RELEVANT COURSEWORK

- **Statistical Analysis:** Applied Regression Analysis, Advanced Probability (Bayesian Networks)
- **Machine Learning:** Deep Learning, Reinforcement Learning, Application of Graph Theory
- **Data-Driven Dynamic System Modeling:** Control Systems

RELEVANT PROJECTS

GPT based time series prediction for real world data

- Replaced TIMEGAN (Time-series Generative Adversarial Networks) with GPT models, resulting in an 84% improvement in prediction accuracy for spatial forecasting of time series data from dynamic chemical sensors.

LLM based video summarizer

- Made a fast online video summarizer using Gemini pro LLM model and Youtube video api.

Data Driven Plume Simulator

- A data-driven fast chemical plume simulator was developed to provide realistic experience for a simulated agent, using gaussian process and Bayesian optimization methods ([PyTorch](#) and Python).

RELEVANT PUBLICATIONS

- Nag Arunava, and Floris Van Breugel. "*Odour source distance is predictable from a time history of odour statistics for large scale outdoor plumes*", [Journal of Royal Society Interface \(2024\)](#)
- Lingenfelter Bryson, Arunava Nag, and Floris van Breugel. "*Insect inspired vision-based velocity estimation through spatial pooling of optic flow during linear motion.*" [Bioinspiration & Biomimetics \(2021\)](#)

PRESENTATIONS

- Invited for participating and speaking at BISCIT, Bio-Inspired Sensing, Computing, and Control with International Teams, 2023 London.
- Presented, “Outdoor odor localization”, and “Data driven plume simulator”, at AI Dynamics Institute workshop, University of Washington, 2023
- Presented, “Mapping outdoor odor plumes using a mobile chemical sensor”, in APS 2022, Chicago